

Cinema Outings AI

Multi-Agent Autonomous Architecture, A2UI Protocol & Cloud Infrastructure

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Submission Track: Google AI L200	Region: us-central1 (Council Bluffs)	Infrastructure as Code: Terraform >= 1.5.0

Executive Summary: Cinema Outings AI is a production-grade, full-stack multi-agent platform designed to discover cinema screenings, personalize recommendations using taste memory, guarantee transaction safety, book tickets, and synchronize calendar invites. Built using the **Google Agent Development Kit (ADK)**, **Gemini 2.5** strategic tiered model routing, **FastMCP** (Model Context Protocol), and Flutter's dynamic **A2UI (Agent-to-UI)** declarative protocol, the architecture is provisioned with enterprise-grade **Terraform** Infrastructure as Code (IaC) on Google Cloud Platform.

1. System Architecture Diagram & Layer Hierarchy

The architecture is partitioned into six decoupled layers operating in continuous synchronization:

1. CLIENT LAYER (Flutter Cross-Platform Mobile & Web)

- Dynamic A2UI Renderer Engine (MovieCard, SeatMap, TicketPass, CalendarInvite, SeenHistory widgets)
- AgentClient with W3C Distributed Tracing (traceparent context propagation header)

▼ HTTP REST / Dynamic A2UI Declarative Protocol JSON

2. FASTAPI GATEWAY & DISTRIBUTED OBSERVABILITY

- Unified Gateway Endpoints: /api/v1/agent/chat, /api/v1/a2ui/action, /api/v1/telemetry
- OpenTelemetry W3C Middleware & Structured JSON Logging (Google Cloud Logging semantic format)

▼ ADK Lifecycle Hooks & Security Defense Pipeline

3. SECURITY GUARDRAILS & STRATEGIC MODEL ROUTING

- InputSecurityGuardrailPlugin:** Intercepts prompt injections, overrides, and malicious payloads
- CloudDlpInspectionService:** Google Cloud DLP PII sanitization (Cards, SSN, Phone, Email, Auth Tokens)
- SemanticRouter:** Real-time intent classification and cost-optimized tiered Gemini model routing
- Evaluation Plugins:** 0% duplicate watched movie enforcement, transaction state machine, cost tracking

▼ Google ADK Supervisor Host Delegation

4. MULTI-AGENT ADK CORE

- Outing Coordinator Agent (Supervisor/Host):** Coordinates memory passing and bundles A2UI payloads (gemini-2.5-flash)
- Search & Reco Agent:** Deep taste grounding (gemini-2.5-pro) & fast catalog search (gemini-2.5-flash)
- Booking Agent:** Zero-temperature deterministic financial & seat hold transaction executor (gemini-2.5-flash)
- Housekeeping Agent:** Watched history archive, favorites memory curation, and calendar invites (gemini-2.0-flash-lite)

▼ External Protocol Execution & Function Invocation

5. TOOLS & PROTOCOL INTEGRATION LAYER

- FastMCP Movie Search Server:** search_cinemas, get_showtimes, get_movie_details, list_now_showing
- Booking Transactions:** get_seat_availability, hold_seats_reservation, process_ticket_payment
- Calendar & Compaction:** RFC-5545 iCalendar (.ics), Google Calendar deep links, and context compaction

▼ Async Non-Blocking Persistence & Google Cloud Infrastructure

6. PERSISTENCE & GOOGLE CLOUD INFRASTRUCTURE (Terraform IaC)

- SQLite Database:** cinema_sessions.db with non-blocking async CRUD & active dialogue compaction
- Google Cloud Run v2:** Auto-scaling backend with /health probes, least-privilege IAM SA, and GCS FUSE volume
- GCP Services:** Artifact Registry, Secret Manager, Cloud DLP, Cloud Trace, and Cloud Monitoring Dashboard

2. Specialized Multi-Agent System Breakdown

Each agent executes a specialized role within clear isolation boundaries, coordinated via shared Google ADK session memory:

Agent Role	Technology & Model	Core Architectural Responsibilities	Session Memory Keys
1. Outing Coordinator (Supervisor/Host)	Google ADK gemini-2.5-flash	• Conversational intent classification • Sub-agent routing & delegation • A2UI declarative widget packaging • Dialog lifecycle & turn history	conversation_history active_intent session_id
2. Search & Reco	Google ADK + FastMCP gemini-2.5-pro (Deep) gemini-2.5-flash (Fast)	• Live screenings lookup via FastMCP • Director & genre taste-matching • 0% duplicate seen movie filtering • Venue, hall & showtime discovery	Reads: favorite_movies seen_movies
3. Booking Agent	Google ADK + Custom Tx gemini-2.5-flash (T=0, Deterministic)	• Real-time seating grid inspection • 10-minute temporary seat reservation • Payment settlement & idempotency • Digital QR token boarding pass	Mutates: active_reservation active_booking
4. Housekeeping	Google ADK + Summarizer gemini-2.0-flash-lite gemini-2.5-flash	• Watched archive ledger maintenance • User favorites list curation • RFC-5545 .ics & Google Calendar export • Automatic conversation compaction	Mutates: seen_movies favorite_movies summary

3. Tools, Protocols & Integration Specifications

Standardized tool contracts connect the multi-agent core with cinema catalogs, transactional APIs, and client widgets:

Protocol / Subsystem	Implementation File	Methods & Interface Signatures	Operational Guarantees
FastMCP Server (Model Context Protocol)	backend/mcp_servers/ movie_search_server.py	• search_cinemas(location, query) • get_showtimes(movie_title, date) • get_movie_details(movie_title) • list_now_showing(genre)	FastMCP stdio server providing screening catalogs, showtimes, auditorium formats (IMAX, standard), and capacity limits.
Ticketing Engine (Custom Transactions)	backend/tools/ booking_transactions.py	• get_seat_availability(c, h, t) • hold_seats_reservation(...) • process_ticket_payment(...) • cancel_booking(booking_id)	Atomic reservation management. Implements lock-lease timeouts (600s), idempotency keys, subtotal calculations, and digital QR boarding passes.
Calendar Engine	backend/tools/ calendar_tools.py	• generate_ics_calendar(...) • create_google_calendar_link(...)	RFC-5545 iCalendar specification generator (`.ics`) and browser-compatible Google Calendar deep links with cinema venue and seat data.
A2UI Protocol (Agent-to-UI)	backend/protocols/ a2ui.py	• A2UIMessage • movie_card, seat_map_selector • ticket_pass, calendar_invite_card • seen_history_list	Declarative JSON protocol enabling autonomous backend agents to construct native Flutter widgets dynamically without app recompilation.
Compaction Engine	backend/state/ history_summarizer.py	• compact_conversation_history(...) • extract_key_entities(...)	Autonomous conversation compaction. Compresses aged dialogue turns into running entity summaries while keeping recent turns intact.

4. Security Guardrails, Cloud DLP & Observability

The system enforces defense-in-depth across the agent execution lifecycle using Google ADK BasePlugin hooks, OpenTelemetry distributed tracing, and Google Cloud Sensitive Data Protection (DLP):

Subsystem	Implementation File	Security & Operational Guarantees
Cloud DLP (Sensitive Data)	backend/security/dlp_service.py	Inspects & redacts sensitive PII before agent processing: Credit Cards ([REDACTED_PAYMENT_CARD]), Emails, Phone Numbers, Social Security Numbers, Bearer/API Tokens.
Input Security Guardrail	backend/plugins/security_guardrails.py	Intercepts prompt injection attempts ('Ignore previous instructions', role hijacking), jailbreak sequences, and malicious formatting attacks before model execution.
Booking Safety Guardrail	backend/plugins/security_guardrails.py	Enforces that payments cannot execute without an active, unexpired seat hold token. Blocks replay attacks, double-charges, and seat reservation concurrency collisions.
OpenTelemetry Tracing	backend/telemetry/tracing.py	W3C traceparent HTTP header propagation across Flutter client, FastAPI gateway, LLM inference, MCP tool calls, and SQLite persistence.
Structured JSON Logging	backend/telemetry/logging.py	GCP Cloud Logging semantic format with severity levels, trace correlation (logging.googleapis.com/trace), span IDs, and session context metadata.

5. Strategic Model Routing & Terraform Infrastructure (IaC)

Semantic intent classification routes queries to optimal Gemini model tiers, all provisioned via modular Terraform:

Model Tier	Model Identifier	Assigned Agents & Roles	Architectural & Cost Rationale
Deep Reasoning	gemini-2.5-pro	Search & Reco (Taste Analysis)	Nuanced stylistic comparisons, director filmography synthesis, and complex multi-criteria recommendation grounding.
Fast Conversational	gemini-2.5-flash	Coordinator, Search (Catalog)	Sub-second intent classification, conversational turn handling, and high-throughput screening discovery.
Deterministic Tx	gemini-2.5-flash (T=0)	Booking Agent	Strict schema compliance, zero-temperature execution for reservation holding and financial transaction calls.
Ultra-Lite	gemini-2.0-flash-lite	Housekeeping & Calendar	Lowest cost tier for conversation compaction, profile memory updates, and simple RFC-5545 `.ics` formatting.

Modular Terraform Infrastructure as Code (GCP):

- **modules/apis:** Enables 14 GCP APIs (Cloud Run, Cloud DLP, Vertex AI, Secret Manager, Cloud Trace, Logging, Monitoring).
- **modules/iam:** Creates dedicated least-privilege service account (cinema-outings-sa) with strict role bindings.
- **modules/secrets:** Manages GEMINI_API_KEY in Google Secret Manager with automated accessor policies.
- **modules/storage:** Provisions encrypted GCS bucket with versioning & lifecycle rules, mounted to Cloud Run via FUSE.
- **modules/registry:** Private Artifact Registry Docker repository for container images.
- **modules/cloud_run:** Cloud Run v2 service deploying the FastAPI multi-agent backend with startup/liveness health probes.
- **modules/monitoring:** Cloud Monitoring dashboard (latencies, CPU/RAM, requests, Cloud DLP PII log metrics) and 5xx alerts.
- **Environments:** Pre-configured environments/dev (cost-optimized) and environments/prod (high-availability warm instances).

Automated Verification & Test Results: All 63 pytest test cases pass cleanly across agent delegation, memory passing, A2UI protocol serialization, MCP tool execution, security guardrails, Cloud DLP sanitization, OpenTelemetry tracing, conversation compaction, and Terraform syntax/module validation.